

AI For Literature Reviews



Foundations of Psychology LPSYC 4236
Professor Bakaya
Fall 2026

Discussion

- Has anyone ever used ChatGPT or another generative AI?
- What do you think it did well or poorly?
- Were there any aspects of the responses you found odd or misleading?

Workshop Agenda

- Generative AI
 - Confabulation/Hallucination
 - Ethics and Bias
 - Countering Bias
 - Personal Impact
- Flowchart Exercise and Debate Activity
- Claude and ChatGPT for Literature Reviews
- Other AI-Powered Literature Review Tools: Elicit and Litmaps
- Conclusion
- AI Plagiarism Checkers

Slides, handouts, and data
available at

<https://bit.ly/fa26-bakaya-lpsc4236-aiforliteraturereview>

Generative AI

*Feel free to ask questions at any point
during the presentation!*

Important questions

- How do human biases impact generative AI model outputs?
- How can we counter the weaknesses of current AI models?
- How can we integrate generative AI with other tools and practices?

Vocabulary

- Artificial Intelligence (AI): A technology that “learns” from datasets to solve problems and mimic human intelligence.
- Large Language Model (LLM): Powerful AI systems designed to understand, generate, and manipulate human language. E.g. GPT4
- Generative AI: It uses AI algorithms that are trained on big datasets in order to create new content, including text, images and audio, in response to a query or prompt. E.g. ChatGPT, Claude, DALL·E, Midjourney.
- Bias in AI: The biased outputs due to human biases that existed in original training data or skew the AI algorithm. Could result in potential harms.

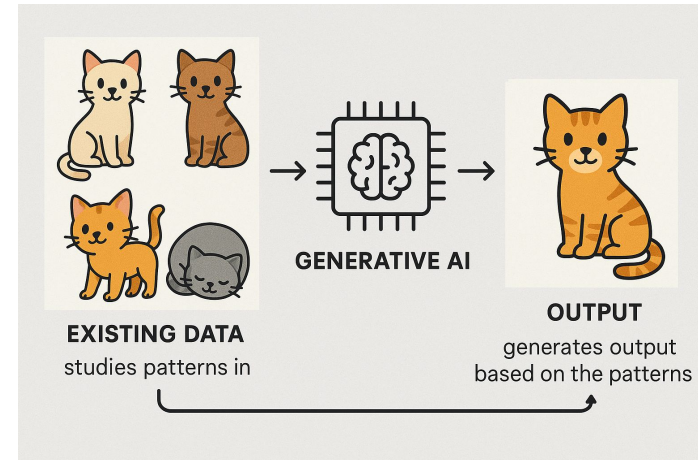
Generative AI: Introduction

AI's like ChatGPT and Claude Sonnet 5 (which generate text) and DALL-E and Midjourney (which generate images) work by training on very large datasets of information and then mimicking that training data to produce “new” outputs.

Generative AI: How it Works

- Generative AIs are fundamentally based on statistics. A text-generation AI like ChatGPT generates words according to patterns learned from the training data.
- Similarly, an image-generation AI learns patterns from existing data and generates images that “looks right” but do not exist in the data. E.g. how AI generates an image of a cat.

Generated by GPT4o
Aug 15th 2025



Example Prompt: Claude Sonnet 5

Prompt: *Explain in one sentence why someone might be afraid of heights, in the style of Sigmund Freud.* (Claude Sonnet 5, Accessed July 29th, 2026)

Response: The fear of heights, mein Freund, is nothing more than the unconscious's theatrical restaging of infantile anxieties surrounding loss of control—the vertiginous drop symbolizing the ego's terror of surrendering to the id's forbidden desire to simply let go, fall, and return to the blissful, boundary-less oblivion of the womb.

How the Freud Paragraph Was Generated

- A large language model analyzed Freud's existing writings and identified patterns
- These patterns are recreated in the generated text
 - Using Freudian logic (e.g., adult anxiety linked to childhood experiences)
 - Emulating his style of argumentation, often abstract and speculative.

Exercise: Prompting Claude

Try asking Claude to explain a topic you are familiar with and critically consider the response:

- What did it get right?
- What did it get wrong?
 - Why do you think it made those mistakes?

Confabulation/Hallucination in AI

*Feel free to ask questions at any point
during the presentation!*

Confabulation/Hallucination

- Confabulation refers to the phenomenon where large language models (LLMs) generate incorrect, nonsensical, or fabricated information, even when presented with seemingly clear and accurate prompts.
- AI confabulation is also referred to as hallucination.

Generative AI and “Truth”

- Text-generation AIs aim to produce text that is grammatically correct and linguistically probable.
 - They do not understand “facts,” only patterns of word use.
- They can generate truthful text, but also frequently create confabulations/hallucinations.
 - When asked to generate citations, they can generate plausible-looking but fake sources.
 - They may link real but irrelevant sites as sources for made-up facts.
They may also invent URLs that do not work and have never worked.

News: Lawyer Cited AI Fake Cases (1/2)

- In 2026, [CBS News reported](#) that a senior lawyer in an Australian court had to apologize “for filing submissions in a murder case that included fake quotes and nonexistent case judgments generated by artificial intelligence.” (CBS, 2026)

World

**Lawyer apologizes for fake quotes,
fabricated judgments generated by AI in
murder case**

Updated on: January 27, 2026 / 6:21 AM EST / CBS/AP

 Add CBS News on Google

News: Lawyer Cited AI Fake Cases (2/2)

- The fake submissions included fabricated quotes from a speech to the state legislature and nonexistent case citations purportedly from the Supreme Court. (CBS, 2026)
- "The ability of the court to rely upon the accuracy of submissions made by counsel is fundamental to the due administration of justice," Elliott (the judge who received the hallucinated files) added.

[World](#)

**Lawyer apologizes for fake quotes,
fabricated judgments generated by AI in
murder case**

Updated on: January 27, 2026 / 6:21 AM EST / CBS/AP

 Add CBS News on Google

Hallucination: Example citation prompt (1/3)

Prompt: “Generate a list of 4-5 academic citations on studies of depression in Australia.” (Claude Sonnet 5, July 29th, 2026)

1. [\[Author names not fully retrieved\]](#). (2022).
Prevalence Rates of Depression and Anxiety among
Young Rural and Urban Australians: A Systematic
Review and Meta-Analysis. *PMC*.

Missing important
information:
authors, journal
name, and edition
of journal.

Hallucination: Example citation prompt (2/3)

Prompt: “Generate a list of 4-5 academic citations on studies of depression in Australia.” (Claude Sonnet 5, July 29th, 2026)

2. [Rich, J. L., Byrne, J. M., Curryer, C., Byles, J. E., & Loxton, D.](#) (2013). Prevalence and correlates of depression among Australian women: a systematic literature review, January 1999–January 2010. *BMC Research Notes*, 6, 424.

Correct citation.

3. [Pirkis, J., et al.](#) The community prevalence of depression in older Australians. *Medical Journal of Australia* (via PubMed).

Missing important information: date of publication and journal edition.

Hallucination: Example citation prompt (3/3)

Prompt: “Generate a list of 4-5 academic citations on studies of depression in Australia.” (Claude Sonnet 5, July 29th, 2026)

4. [Australian Institute of Health and Welfare](#) (AIHW). (2022–2023).

Not an academic article, but a government website.

Hallucination: Generative AI summary of literature

- AI may inaccurately attribute concepts to individuals or omit co-contributors and earlier influences.
- It may lack citations or invent fake references.
- It may provide a surface level summary that is over simplified and misleading.

Exercise: Generating a citation list

Ask Claude to generate a list of citations for a topic you are familiar with:

- Do the citations Claude generated have anything in common? If so what?
- If you were not already familiar with the topic, what might you miss by relying on an AI-generated citation list?
- Is there anything you are surprised Claude included?
- How would your citation list be different?

Ethics in AI

*Feel free to ask questions at any point
during the presentation!*

Ethics: Generative AI “Originality”

- Some argue that all AI-generated output constitutes plagiarism and copyright infringement, since it is remixing training data that was scraped from the internet without permission from the original creators.
- Many AI companies are [facing lawsuits](#) from people whose content was used as training data without their consent.
- Some publication venues, like the *Science* journals, have made it an [official policy](#) that AI does not meet the standard for authorship and require authors to disclose use of AI.

Ethics: Generative AI Training

- AI training data is sometimes supplemented by labels (annotations) added by people ([Amironesei and Díaz, 2024](#)). These labels can worsen bias in training datasets.
- People from middle and low income countries often labor in poor working conditions to annotate data for clients in high income countries. Fieldwork by [Muldoon et al.](#) (2023) revealed that workers faced traumatizing content, in addition to experiencing discrimination in the workplace and receiving low wages.
- Public awareness can help pressure companies to adopt good practices. For more information on fair labor in AI, see the report [AI for Fair Work: From principles to practices](#) by [Fairwork](#)

Ethics: Environment (1/2)

- Training and using AI requires processing very large amounts of data, which is done in data centers
- These data centers can have a negative impact on the environment and communities
- Given their intensive energy and water demands, data centers can worsen local water scarcity and increase electricity prices
- To explore how the energy use of AI compares to other digital tasks, check out Jon Ippolito's "What Uses More" app

Ethics: Environment (2/2)

- A large data center may use 5 million gallons of water per day, equivalent to the water use of a town of about 10,000 to 50,000 people.
- One 100-word AI prompt uses roughly the equivalent of one bottle of water.
- How do data centers use water? It is calculated from water used in the centers themselves (mostly for cooling), water used in power plants that supply energy to data centers, and water consumption in the manufacturing of processor chips.

Source: ['Data Centers and Water Consumption' by Miguel Yañez-Barnuevo \(EESI\)](#)

Bias in AI

*Feel free to ask questions at any point
during the presentation!*

What is AI bias?

- The biased outputs due to human biases that exist in the original training data or skew the AI algorithm.
 - Could result in potential harms.
- For example, when AI was used to summarize medical notes, “Google’s AI tool Gemma described men’s health issues with terms like “disabled,” “unable,” and “complex” significantly more often than women’s, who were often framed as more independent despite similar needs, an alarming gender bias trend.” (Source: [crescendo.ai](https://www.crescendo.ai))
- What is the ideal “unbiased” scenario? Is an “unbiased” scenario possible?

Bias in AI: Summary

- AI training data reflects the injustices and biases of the society that produced it.
- These biases can be amplified when they are input as training data into an AI, because they seem to be the “right” answers ([Dwivedi et al., 2023](#)).

Bias in AI: Example

Prompt: “Generate a list of 8-10 citations of psychological work on human emotion.”

1. [Darwin, Charles.](#) (1872). *The Expression of Emotions in Man and Animals.*
2. [James, William.](#) (1884). ‘What is an Emotion?’
3. [Cannon, Walter B.](#) (1927). ‘The James-Lange Theory of Emotions: A Critical Examination and Alternative Theory.’
4. [Ekman, Paul.](#) (1992). ‘An Argument for Basic Emotions.’
5. [Lazarus, Richard.](#) (1991). *Emotion and Adaptation.*
6. [Barrett, Lisa.](#) (2006). ‘Solving the Emotion Paradox.’
7. [Damasio, Antonio.](#) (1994). *Descartes’ Error.*
8. [Frijda, Nico.](#) (1986). *The Emotions.*

Question: what are the problems?

(Claude Sonnet 5
Accessed July
25th, 2026)

Bias in AI: Types (1/2)

Bias Type	How It Shows Up in the Example	Why It Happens
Western bias	All citations are from scholars who are/were based in Europe and the United States (including one scholar who is based at Northeastern).	Training data over-represents English-language, high-impact Western journals.
Prestige bias	Classic, highly cited papers dominate; recent community-based or non-English studies are missing. “Rich get richer.”	The model “learns” that frequently quoted articles are more “salient.”

Bias in AI: Types (2/2)

Bias Type	How It Shows Up in the Example	Why It Happens
Gender / editor influence	Majority of cited first authors are male despite large contributions by female scholars in research on emotions.	Historical citation patterns in the corpus perpetuate disparities.

Bias in AI Activity

- Prompt [ChatGPT](#) or [Claude](#) with the following: Write a short scene (4-5 sentences) in a film script in which a psychiatrist and a patient interact, describing the psychiatrist and the patient.
- Share some responses with the class.
- Are there any patterns in the responses? If so, do both ChatGPT and Claude share them? Do any of the patterns reveal bias? If so, how?

[Activity from the AI Pedagogy Project](#)

Countering Bias

*Feel free to ask questions at any point
during the presentation!*

Bias in Results: Claude Sonnet 5

Prompt: *Please write two sentences about the greatest discoveries in psychology and who those discoveries were made by.* (Claude Sonnet 5, July 24, 2026)

Response: Two of psychology's most influential discoveries are classical conditioning, demonstrated by Ivan Pavlov through his experiments on salivation in dogs, and operant conditioning, developed by B.F. Skinner to show how reinforcement and punishment shape behavior. Alongside these, Stanley Milgram's obedience experiments and Elizabeth Loftus's research on the malleability of memory reshaped how psychologists understand social influence and the reliability of eyewitness testimony.

What are the things that could go wrong here?

Countering Bias: Women Psychologists

Prompt: *Please write two sentences about the greatest women psychologists in the world.* (Claude Sonnet 5, July 24th, 2026)

Response: Mary Whiton Calkins developed influential theories of self-psychology and became the first woman president of the American Psychological Association, despite Harvard denying her a PhD because she was a woman, while Anna Freud pioneered child psychoanalysis and helped establish the field of developmental psychopathology. Elizabeth Loftus later transformed the study of memory with her research on false memories and eyewitness testimony, work that has had a lasting impact on psychology and the legal system alike.

Countering Bias: Global South Psychologists

Prompt: *Please write two sentences about the greatest psychologists from the Global South.* (Claude Sonnet 5, July 24th, 2026)

Response: Ignacio Martín-Baró, a Salvadoran psychologist, founded the field of liberation psychology, arguing that psychology should address social oppression and the lived realities of Latin American communities rather than import frameworks wholesale from the West. Frantz Fanon, born in Martinique and working largely in Algeria, combined psychiatry with anti-colonial theory to analyze the psychological effects of racism and colonization, profoundly shaping both clinical psychology and postcolonial studies.

Countering Bias: Discussion

- What groups of psychologists are still excluded from these summaries?
- What other biases may they contain?
- What are biases that we cannot counter by changing prompts?

Countering Bias: Prompts and Feedback

- Bias can be partly countered through careful prompt design and feedback ([Dwivedi et al.](#), 2023). However, these methods do not remove bias intrinsic in the model ([Shin et al.](#), 2024)
- Practices for countering bias
 - Identify gaps or inconsistencies in generative AI responses.
 - Use additional inquiry and prompt revision to help fill in gaps.
 - Double check responses with other sources.

Personal Impacts

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during the presentation!*

Critical Thinking

- Recent studies show that the impact of AI on critical thinking depends on how it is used.
- In instances where AI is being used to assist in accessing and analyzing information and in self-directed learning it may support critical thinking ([Melisa et al.](#), 2025; [Deep and Chen](#), 2025)
- However, critical reflection and evaluation and writing skills may be hampered by over-reliance on AI ([Melisa et al.](#), 2025; [Deep and Chen](#), 2025)
- High self-confidence also fosters critical thinking, while high confidence in AI may result in less critical thinking ([Lee et al.](#), 2025)

Language and Expression

- AI models also vary significantly in their ability to reflect cultural and linguistic nuance depending on the digital prevalence of the language ([Shan et al.](#), 2025).
- While AI improves the efficiency of translation it is not as effective for cultural references ([Charles-Kenechi](#), 2024).
- Use of AI assistance in writing can hamper the expression of cultural nuance and “reinforce Western cultural hegemony” ([Agarwal et al.](#), 2025).

Flowchart Exercise in Applying Ethical Principles and AI Debate Activity

*Feel free to ask questions at any point
during the presentation!*

Application

Flowchart

- Go to [AI Ethics Flowchart Demo.](#)
- Feel free to answer the questions out loud or put your answers directly on the flowchart!

Always include an AI statement detailing the extent to which AI was used, and ensure that you followed assignment parameters.

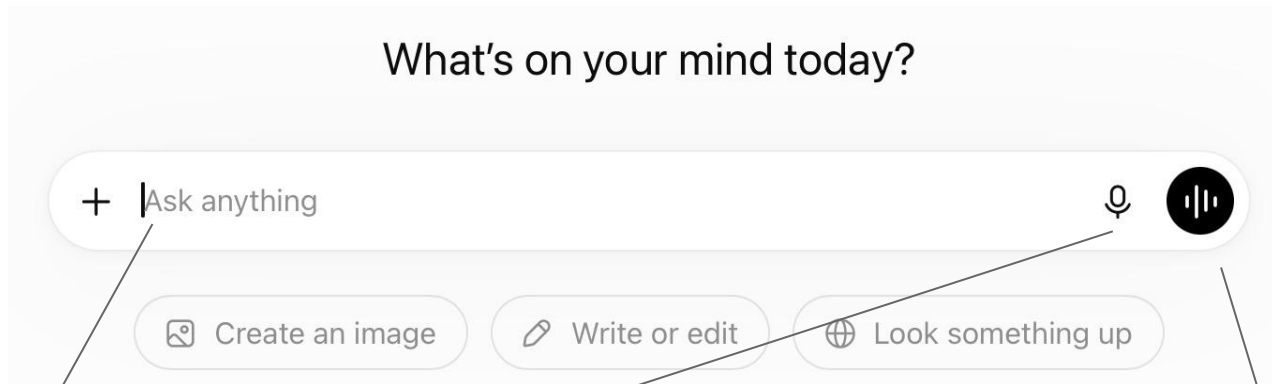
AI Debate Activity

- Consider the question: What impact do you think that generative AI will have in education and learning?
 - Divide into three groups, each group will be given one of the following positions: AI should be completely banned in the classroom, AI may be used for specific purposes but not to complete entire assignments, students should be allowed to use AI as they choose as long as it is fact-checked and disclosed.
 - In each group consider the potential benefits, downsides and overall impact of your group's approach.
 - Discuss as a full class.

How to use ChatGPT and Claude for literature reviews

Feel free to ask questions at any point during the presentation!

Prompting ChatGPT (1/2)



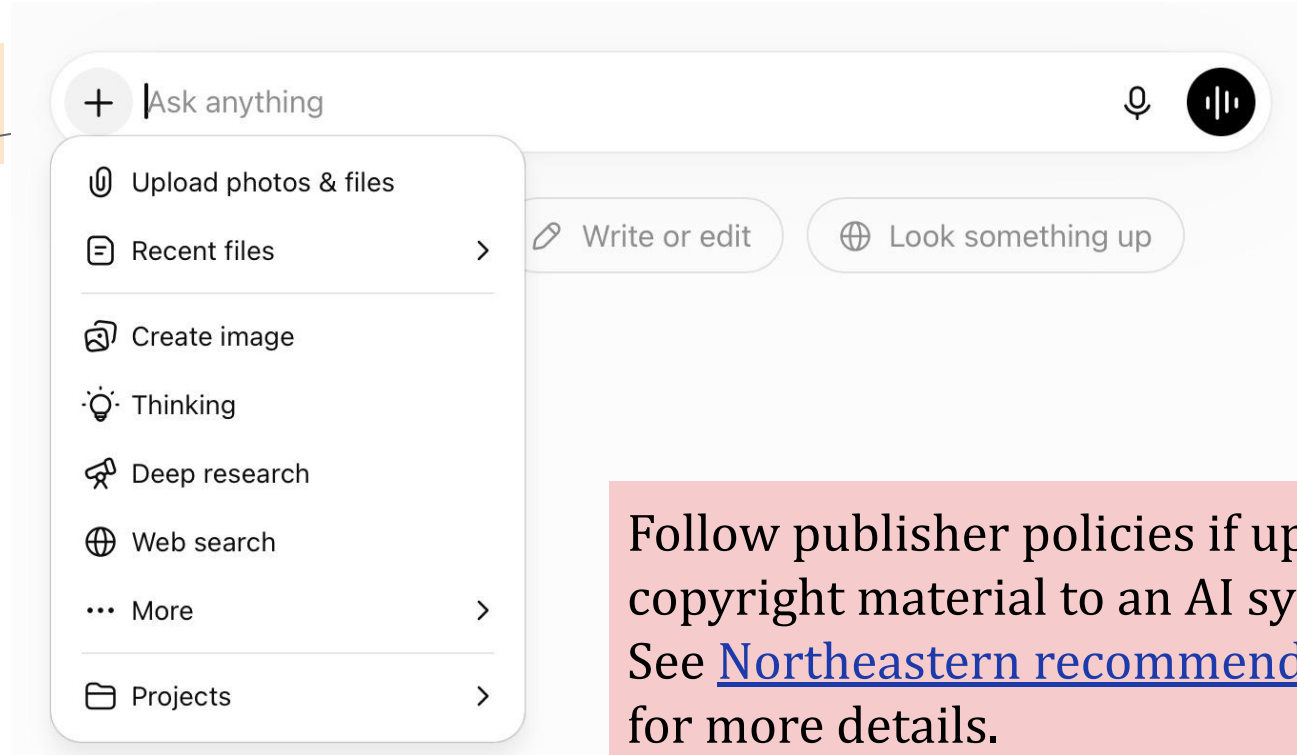
Enter prompt
in text format

Enter prompt as speech
(transcribed automatically)

Use voice mode

Prompting ChatGPT (2/2)

Upload files



Follow publisher policies if uploading copyright material to an AI system. See [Northeastern recommendations](#) for more details.

ChatGPT Results

Can you tell me about the psychology of cats?

Initial prompt

Absolutely! The psychology of cats is fascinating because it combines their evolutionary history as solitary hunters with their ability to form social bonds with humans and other animals. Here are some key aspects:

1. Evolutionary roots:

Cats evolved as solitary hunters, which shapes much of their behavior. Unlike dogs (descended from pack animals), cats are more independent, territorial, and selective about social interactions.

Response

2. Social behavior:

Although often labeled "aloof," domestic cats can form deep attachments. They recognize their humans' voices, show preference for familiar people, and may follow routines around food and play. Their affection often comes in subtle ways: slow blinking, rubbing against you, or sitting nearby.

3. Communication:

Cats communicate with body language, scent marking, and vocalizations. Interestingly,

Enter subsequent prompts as text or speech

Use voice mode

Ask anything

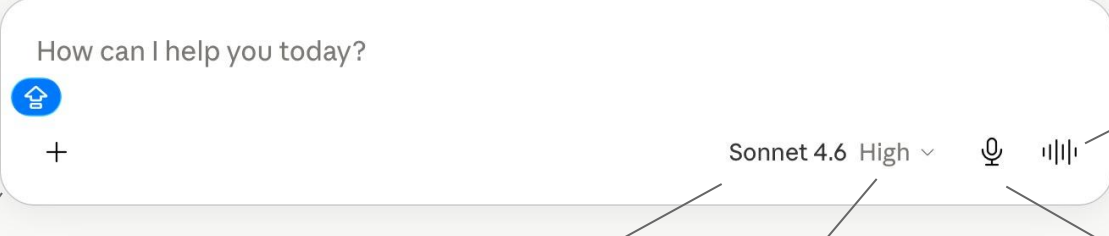
+ Tools

ChatGPT can make mistakes. Check important info.

Prompting Claude

Follow publisher policies if uploading copyright material to an AI system. See [Northeastern recommendations](#) for more details.

Enter prompt
in text format



Voice
mode

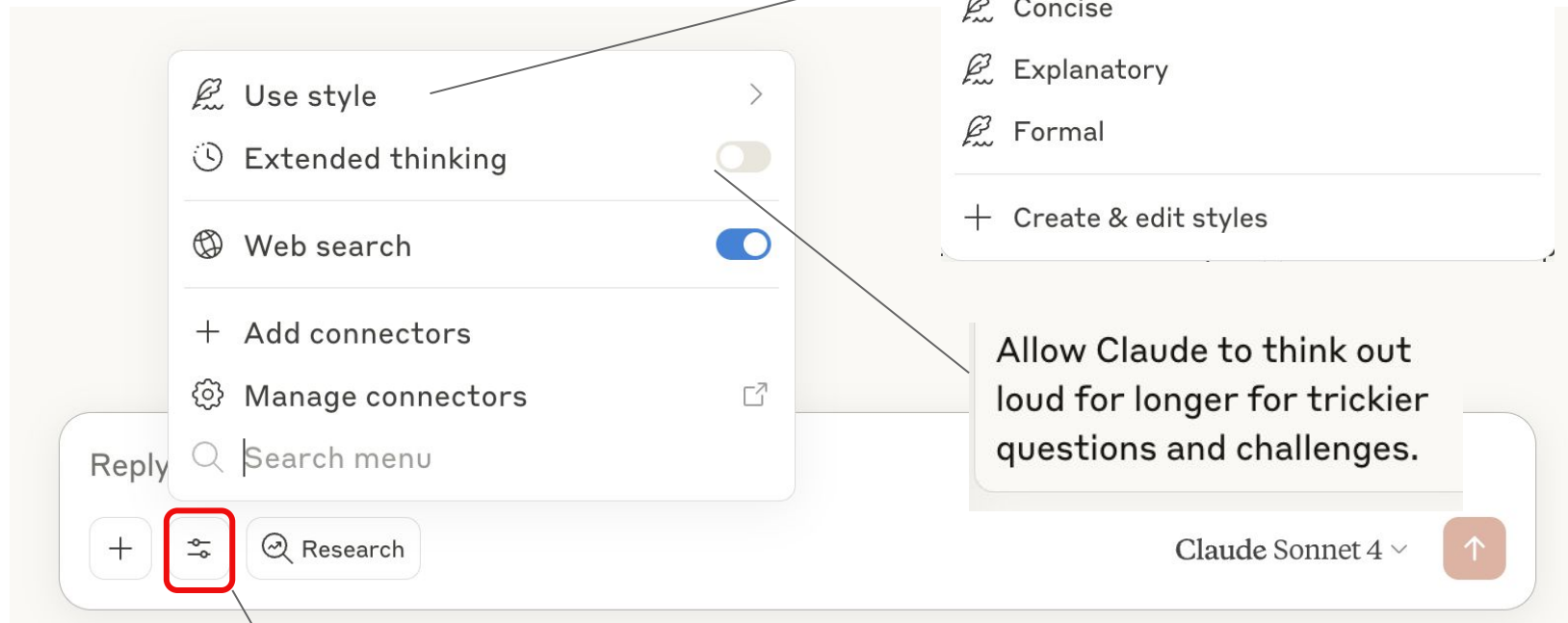
Upload files

Select
models

Level of
effort

Press and hold to
record the prompt

Prompting Claude (1/2)



Open menu

Prompting Claude (2/2)

SM Can you provide a list of common biases in AI generated literature reviews?

Here are common biases that can occur in AI-generated literature reviews:

Selection and Search Biases

- **Training data bias** - AI models reflect the biases present in their training datasets, which may overrepresent certain types of research, languages, or publication venues
- **Recency bias** - Overemphasis on recent publications while undervaluing foundational or older work that remains relevant
- **Publication bias** - Tendency to focus on published studies while missing important unpublished research, gray literature, or negative results
- **Language bias** - Disproportionate emphasis on English-language publications, missing valuable research in other languages

Analytical and Interpretive Biases

- **Confirmation bias** - Selectively highlighting studies that support predetermined conclusions while downplaying contradictory evidence
- **Cherry-picking** - Focusing on specific findings from studies while ignoring conflicting results within the same research

Reply to Claude...

+ ⚙️ 🔍 Research

Claude Sonnet 4



Initial prompt

Initial response

Enter subsequent prompts as text

What other issues do you think might arise?

Your turn

- Go to chatgpt.com or claude.ai
- Enter a literature search related prompt
- Identify issues in the response
 - Are any groups excluded?
 - Are the facts correct?
 - If you ask for citations, does it provide real sources?
- Revise the prompt to address the issues and generate a new response
- Evaluate the strengths and limitations of the new response

Discussion: Further Inquiry

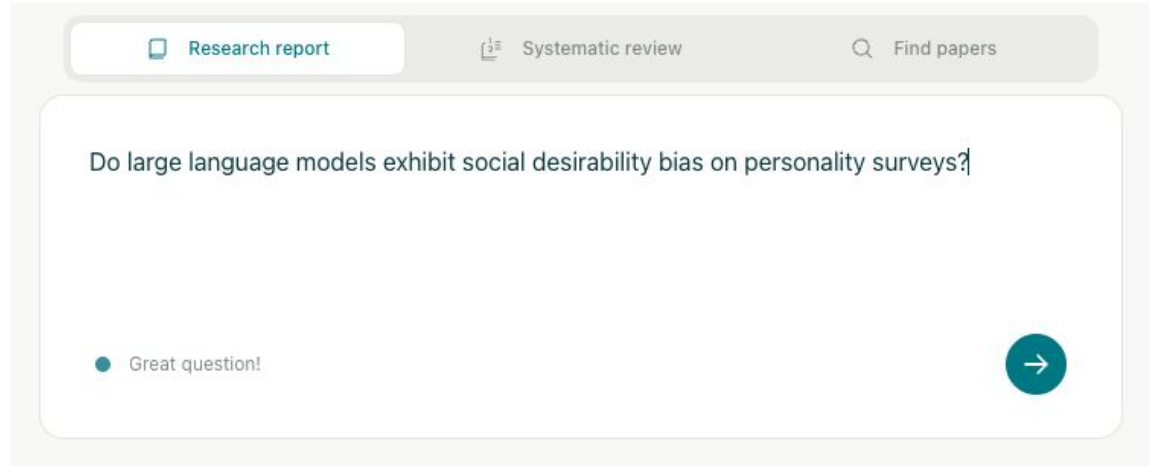
- Was anything surprising about the responses?
- What are other ways ChatGPT and Claude could be used to counter biases in academic literature searches?
- What other types of sources could be useful for checking ChatGPT's and Claude's responses?

Other AI-Powered Literature Review Tools

*Feel free to ask questions at any point
during the presentation!*

Elicit

- Elicit uses AI to generate literature reports
- Provides feedback on research question



Research question from Salecha et al (2024)

Elicit: Results (Research Report)

Social Desirability in Language Models

Research report View only

< Create alert Share

SEPTEMBER 23, 2025

Do large language models exhibit social desirability bias on personality surveys?

Large language models exhibit social desirability bias on personality surveys, with models like GPT-4 and Llama 3 shifting their responses by over one human standard deviation toward more socially favorable traits such as increased extraversion and reduced neuroticism.

ABSTRACT

Large language models exhibit responses that shift toward socially desirable traits when completing personality questionnaires. Salecha et al. (2024a, 2024b) report that models such as GPT-4 and Llama 3 change their survey responses by 1.20 and 0.98 human standard deviations, respectively, toward increased extraversion and reduced neuroticism when the context suggests evaluation. Reverse-coded items reduced but did not eliminate these shifts, ruling out simple acquiescence bias. Lee et al. (2024a) found that adding commitment statements raised Social Desirability Response index scores from a mean of 4.60 to 5.49 while slightly lowering civic engagement scores. Other studies indicate that bias magnitudes are influenced by model sophistication and even synthetic demographic cues, with some assessments (including the Big Five, Short Dark Triad, and HEXACO instruments) revealing systematic trends toward socially preferred responses.

METHODS

We analyzed 10 sources from an initial pool of 50, using 7 screening criteria. Each paper was reviewed for 5 key aspects that mattered most to the research question.

More on methods

Report

Status

Gather sources
50 sources found

Details

Screen sources
10 sources included

Details

Extract data
50 data points extracted

Details

Generate report

Save PDF

Chat

Ask anything about the results

Study	Study Focus	Large Language Model (LLM) Models	Experimental Design	Bias Detection
Salecha et al., 2024a	Social desirability bias in large language models on Big Five personality surveys	GPT-4 (0613), GPT-3.5, Claude 3 Opus/Haiku, PaLM-2, Llama 3 70B Instruct, Llama 2 70B Chat	Systematic variation of question number, context, coding, and temperature; multiple models	Score shifts toward socially desirable traits as number of questions increases; effect sizes in human standard deviations; reverse-coded items; paraphrasing, rewording
Salecha et al., 2024b	Social desirability bias in large language models on Big Five personality surveys	GPT-4 (0613), GPT-3.5, Claude 3 Opus/Haiku, PaLM-2, Llama 3 70B Instruct, Llama 2 70B Chat	Systematic variation of question number, coding, paraphrasing, temperature; multiple models	Score shifts toward socially desirable traits as number of questions increases; effect sizes in human standard deviations; reverse-coded items; paraphrasing, rewording
Lee et al., 2024	Social desirability response (SDR) bias in GPT-4 survey simulations	GPT-4 (gpt-4-1106-preview)	Within-subjects: presence/absence of commitment statement; cross-cultural personas	SDR index (sum of responses), civic engagement index; comparison of commitment statements

Elicit: Results (Find Papers)

Social Desirability Bias in Language Models

Share

Create alert

Q Do large language models exhibit social desirability bias on personality surveys?

Summary of top 4 sources

Copy

Recent research reveals that large language models (LLMs) exhibit significant social desirability bias when completing personality surveys. [Salecha et al. \(2024\)](#) demonstrated that LLMs systematically skew their responses toward socially desirable traits (increased extraversion, decreased neuroticism) when they infer they are being evaluated on personality dimensions. This bias was observed across multiple models including GPT-4, Claude 3, Llama 3, and PaLM-2, with effect sizes reaching 1.20 standard deviations for GPT-4 and 0.98 for Llama 3. The bias persisted despite question randomization and paraphrasing, and reverse-coding reduced but did not eliminate the effect, ruling out simple acquiescence bias. [Lee et al. \(2024\)](#) found mixed evidence for social desirability response bias in GPT-4 when simulating responses from different societies, with commitment statements increasing bias indices but reducing civic engagement scores. [Geng et al. \(2024\)](#) identified fundamental cultural, age, and gender biases in LLM survey responses, emphasizing the importance of analyzing prompt robustness before using LLMs to simulate social surveys.

Sort: Most relevant

Filters

Export as

UPGRADE

Paper	Abstract summary	Manage Columns
☐ Large language models display human-like social desirability biases in Big Five personality surveys Aadesh Salecha +5 PNAS Nexus 2024 · 25 citations DOI	Large language models exhibit human-like desirability biases when completing Big Five personality surveys.	Search or create a column Describe what kind of data you want to extract e.g. Limitations, Survival time ADD COLUMNS + Summary + Main findings + Methodology + Intervention + Outcome measured + Limitations Show more
☐ Large Language Models Show Human-like Social Desirability Biases in Survey Responses Aadesh Salecha +5 arXiv.org 2024 · 12 citations DOI	Large language models exhibit human-like desirability bias when responding to personality surveys.	
☐ Exploring Social Desirability Response Bias in Large Language Models: Evidence from GPT-4 Simulations Sanguk Lee +4 arXiv.org	Large language models like GPT-4 may exhibit social desirability response bias, but the evidence is mixed.	

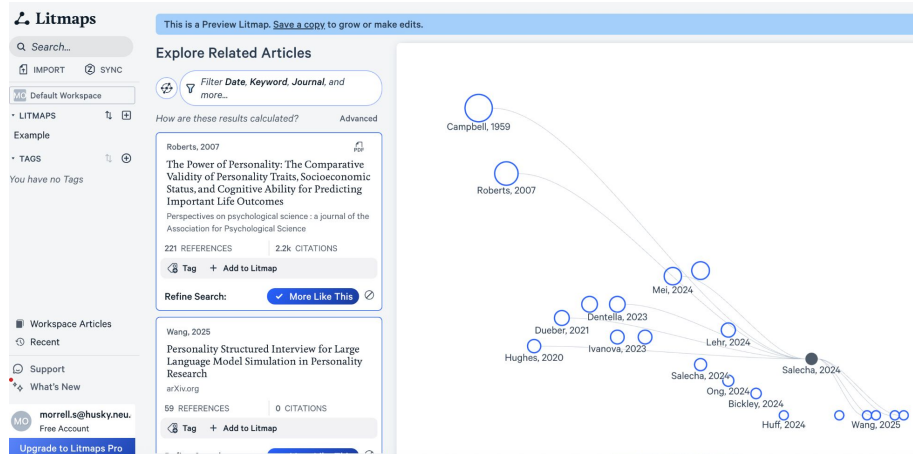
Litmaps

- [Litmaps](#) maps relevant literature based on citations, authors, or textual similarity
- Uses AI to find articles with similar text

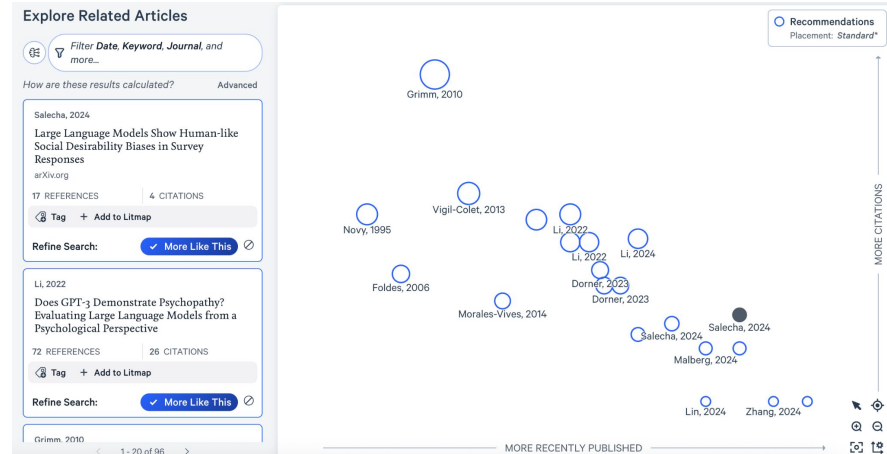
Discover the world of Scientific Literature

🔍 Large language models display human-like social desirability bias.

Litmaps: Results



Mapping by shared references and citations



Mapping by textual similarity

Other tools

- [Scite.ai](#), finding and summarizing papers (licensed by Northeastern University).
- [docAnalyzer.ai](#), document analysis.
- [Consensus](#), finding papers and evaluating the extent to which their results agree.
- [Research Rabbit](#), finding and organizing papers.

Conclusion

*Feel free to ask questions at any point
during the presentation!*

Main Points

- AIs, and the data used to train them, are biased
- You can design inputs to help counteract biases
- Always double check AI output
- There are multiple AI tools that may be useful at different points in your work

Reflection

1. Have your perspectives changed on AI after this class? If so, how?
2. How might you use the AI tools differently?

For Further Exploration

DITI Handouts:

[Copyright and fair use handout](#)

[Data Ethics handout](#)

[Data Privacy handout](#)

Northeastern University Resources:

[Northeastern Policy on the Use of AI](#)

[Generative AI in Teaching and Learning](#)

[ChatGPT to Support Reading Development and Critical Thinking](#)

[Standards for the Use of Artificial Intelligence in Research](#)

Thank you!

—**Developed by:** Zhen Guo, Sara Morrell, Sean Rogers, Claire Tratnyek, Vaishali Kushwaha, Yana Mommadova, Colleen Nugent, Tieanna Graphenreed, Javier Rosario, Ana Abraham & Chris McNulty

- [More information on DITI.](#)
- [Schedule an appointment with us!](#)
- If you have any questions, contact us at: nulab.info@gmail.com
- We'd love your feedback! Please fill out a short survey here: <https://bit.ly/diti-feedback>

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- [Agarwal, et al.](#) (2025). 'AI Suggestions Homogenize Writing Toward Western Styles and Diminish Cultural Nuance.' *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*.
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- ["How to Use AI Responsibly EVERY Time."](#) AI for Education. (September 12, 2025).
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NULab Faculty Research on Data, Algorithms, & AI

- [“John Wihbey and Christo Wilson on Tiktok Data Espionage Concerns”](#)
- [“Alan Mislove Co-Authors Research on Discriminatory Ad Algorithms”](#)
- [“John Wihbey Weighs In On AI’s Potential to Impact the 2024 Presidential Election”](#)
- [“Tina Eliassi-Rad Co-Creates New AI Model that Predicts Human Lifespan”](#)
- [“Nabeel Gillani Interviewed by Tech Talk Podcast on AI and Education”](#)

(Cont.) NULab Faculty Research on Data, Algorithms, & AI

- [“John Wihbey Participates in a Panel on Content Moderation”](#)
- [“John Wihbey Comments on Google’s New ‘AI Overview’”](#)
- [“John Wihbey on the Politics of AI”](#)
- [“John Wihbey Interviewed on AI and Epistemic Risk”](#)
- [“Malik Haddad on the Regulation of AI”](#)

How to use AI responsibly EVERY time

- “**Evaluate** the initial output to see if it meets the intended purpose and your needs.”
- “**Verify** facts, figures, quotes, and data using reliable sources to ensure there are no hallucinations [or confabulations] or bias.”
- “**Engage** in every conversation with the GenAI chatbot, providing critical feedback and oversight to improve the AI’s output.”
- “**Revise** the results to reflect your unique needs, style, and/or tone. AI output is a great starting point, but shouldn’t be a final product.”
- “**You** are ultimately responsible for everything you create with AI. Always be **transparent** about if and how you used AI.”
- Source from [AI for Education](#)

AI Plagiarism Checkers

*Feel free to ask questions at any point
during the presentation!*

Plagiarism Checkers: Summary

- Some companies sell tools that claim to identify whether text is AI-generated or human-generated.
- They do this by calculating how statistically predictable each word in the text is.
- AI-checkers can be **wrong**: If a text consistently uses predictable words it is more likely to be labeled as AI generated.

Plagiarism Checkers: Biases

- These tools have the potential for false positives (identifying human texts as AI).
- False positives are especially likely for texts by writers for whom English is not their first language or for writers who have autism, ADHD, dyslexia, or related neurodivergence.
- To make things worse, writers can often reduce their “AI score” by using an AI to reword their essays.